

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Reverse Engineering Task

For each task, study the described big data solution, infer how it was built, identify the underlying components, and explain what design decisions were likely made.

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing

structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy, encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.

4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.
5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.

Reverse Engineering Task Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Inference Accuracy	25%	Infers system components and flow with high accuracy	Mostly accurate inference	Basic inference	Weak inference
Understanding of Architecture	25%	Shows clear understanding of underlying big data architecture	Good understanding	Partial understanding	Poor understanding
Use of Technical Evidence	20%	Uses strong reasoning and technical evidence	Reasonable evidence	Limited evidence	No meaningful evidence
Depth of Analysis	15%	Analysis is deep and insightful	Reasonably deep	Basic depth	Superficial analysis
Clarity	15%	Clear and organized explanation	Generally clear	Somewhat unclear	Poorly presented

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.

Answer:

- The platform handles billions of website clicks, mobile interactions, and customer transactions. It would likely use HDFS or cloud storage such as Amazon S3 for distributed storage. Apache Kafka can be used to collect and transport data from websites, mobile applications, and transaction systems.
 - Structured data such as customer and transaction records can be stored in relational databases, while semi-structured data such as clickstream logs can be stored in JSON, Avro, or Parquet format. Data can be partitioned based on date, region, or event type to improve query performance.
 - Apache Spark can be used for large-scale batch processing, while Spark Streaming or Apache Flink can process real-time events. Replication and horizontal scaling help the system handle failures and increasing workloads.
 - Finally, processed data can be stored in a data warehouse and connected to Power BI or Tableau for dashboards showing sales, customer behavior, and other business KPIs.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.

Answer:

- An e-commerce platform generates data from product views, cart additions, abandoned checkouts, and payments. Apache Kafka would likely be used to stream these events in real time. Customer profiles can be stored using distributed databases such as MongoDB, Cassandra, or HBase.
- Redis can be used as a caching system to provide fast access to frequently used customer and product information. Clickstream, session, and transaction data can be synchronized using customer ID, session ID, and timestamps.

- Apache Spark or Flink can process the data, while Spark SQL, Trino, or Presto can perform distributed queries. Machine learning models can analyze customer behavior and generate personalized product recommendations.
 - Structured order data can be combined with semi-structured browsing data for customer analytics. Kafka replication, database replication, and partitioning provide fault tolerance and scalability. Power BI or Tableau can display KPIs such as sales, conversion rate, and cart abandonment.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy, encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.

Answer:

- The healthcare system manages patient histories, medical images, wearable sensor data, and prescription records. A hybrid database architecture would be suitable. Relational databases can store structured information such as patient records and prescriptions, while NoSQL databases and cloud storage can store documents, sensor data, and medical images.
 - HL7 and FHIR standards can be used to exchange healthcare information between hospitals, laboratories, and insurance systems. Apache Kafka can collect continuous wearable-device data, while Spark Streaming or Flink can process it in real time.
 - Machine learning models can be developed using historical patient data for disease prediction, risk assessment, and patient monitoring. Apache Spark can perform large-scale population health analysis.
 - Because healthcare information is sensitive, the system should use encryption, authentication, role-based access control, and audit logs. Analytics dashboards can provide doctors and researchers with patient trends, risk scores, and population-level insights.
4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.

Answer:

- A smart city platform collects data from traffic sensors, CCTV systems, weather stations, and public transport. Edge computing can process some data close to the sensors before sending it to the central system. This reduces network traffic and improves response time.
 - Apache Kafka or MQTT can be used for continuous data transmission. Historical information can be stored in HDFS or cloud storage. Time-series databases such as InfluxDB or TimescaleDB can store sensor readings, while PostGIS can support geographical and location-based queries.
 - Spark Streaming or Apache Flink can analyze traffic data in real time and predict congestion. Batch processing can analyze historical traffic patterns for urban planning.
 - Security can be provided through encryption, authentication, role-based access control, and audit logging. Tools such as Power BI, Tableau, or Grafana can provide live dashboards showing traffic, weather, transport activity, and incidents.
5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.

Answer:

- This system combines patient records, wearable readings, laboratory reports, and appointment histories. A hybrid storage architecture can use relational databases for structured medical records, NoSQL databases for flexible data, and a secure data lake for large-scale storage.
- ETL pipelines can extract data from different hospitals and clinics, transform it into a common format, and load it into the analytical system. HL7 and FHIR can help different healthcare systems exchange information.
- Wearable-device data can be processed in real time using Kafka, Spark Streaming, or Flink. Apache Spark can process millions of patient records for population-level healthcare analysis.
- Machine learning models can be used for disease-risk prediction, diagnosis support, patient deterioration prediction, and treatment recommendations. Healthcare data must be protected using encryption, authentication, access control, and audit mechanisms. Dashboards can provide doctors with patient trends and risk scores to support clinical decisions.